Fiscal Year:	FY 2017	**Task Last Updated:**	FY 06/19/2017
PI Name:	DeLucas, Lawrence J. O.D., Ph.D.		
Project Title:	The Effect of Macromolecular Transport on Microgravity Protein Crystallization		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	BIOPHYSICS--Biophysics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Lawrence.delucas@aero.org	**Fax:**	FY
PI Organization Type:	NON-PROFIT	**Phone:**	3103365000
Organization Name:	The Aerospace Corporation		
PI Address 1:	Center for Biophysical Sciences and Engineering		
PI Address 2:	2310 E El Segundo Blvd		
PI Web Page:			
City:	El	**State:**	CA
Zip Code:	90245	**Congressional District:**	33
Comments:	NOTE: PI retired in 2017 from the University of Alabama, Birmingham, and is now Principal Scientist at The Aerospace Corporation, El Segundo, CA (Ed., 5/22/2018)		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2013 Complex Fluids & Macromolecular Biophysics NNH13ZTT001N
Start Date:	08/20/2013	**End Date:**	05/14/2017
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:	ISS		
Flight Assignment:	ISS NOTE: End date changed to 5/14/2017 (original end date was 8/19/2018) due to PI move to The Aerospace Corporation (Ed., 5/22/18)		
Key Personnel Changes/Previous PI:	September 2016: none.		
COI Name (Institution):	Betzel, Christian D.Sc. (University of Hamburg) Dierks, Karsten Ph.D. (University of Hamburg) Meyer, Arne Ph.D. (University of Hamburg)		
Grant/Contract No.:	NNX13AQ23G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Proteins are important macromolecules without which our bodies would be unable to repair, regulate, or protect themselves. Determination of the atomic three-dimensional structure of proteins provides critical information that allows scientists to understand how they function. It also provides critical structural information to support the development of new pharmaceuticals to treat chronic and infectious diseases. X-ray crystallography is used to determine protein structure but first requires the growth of well-ordered protein crystals that are of sufficient quality. Protein crystals previously grown in microgravity have resulted in more than fifty examples of clear improvements in crystal quality via x-ray diffraction analysis. The majority of the crystal improvements resulted in improved resolution of the three-dimensional protein structures as determined via x-ray crystallography. Two theories have been proposed to explain the observed microgravity crystal improvements. Both are based on protein transport rate differences in microgravity versus unit gravity environments. One theory is based on expected slower crystal growth rates, due to slower protein transport to the growing crystal surface. The second theory proposes that there is an exclusion of larger protein aggregates becoming incorporated into the growing crystal (aggregates produce crystal defects that reduce crystal quality). The theoretical explanation is based on the expected faster transport of single protein molecules as opposed to aggregates of two or more protein molecules. Unfortunately, there is insufficient experimental data to validate either theory. This proposal will utilize different Light Microscopy Module (LMM) optical systems to determine which theory, via a series of experiments, addresses the void in our understanding of microgravity's effect on growing protein crystals. NOTE: PI retired in 2017 from the University of Alabama, Birmingham, and is now Principal Scientist at The Aerospace Corporation, El Segundo, CA; project continues with same title as grant 80NSSC18K0013 (Ed., 5/22/2018)
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The ultimate goal of this research is to determine the mechanism by which microgravity affects the quality of protein crystals. The research will utilize The Light Microscopy Module (LMM) to investigate two possible theories for the improved crystal quality observed in a microgravity environment: 1) Slower crystal growth rates result in protein that is more perfectly ordered in the crystalline lattice and/or 2) a microgravity environment basically filters out larger aggregates of protein thereby allowing a more uniform population of single protein molecules to become incorporated into the crystalline lattice, thereby producing higher quality crystals. Understanding the mechanism by which microgravity produces higher quality crystals can lead to new ideas for Earth-based crystallization systems that may also improve the quality of protein crystals. This will benefit the structural biology community that determines protein structures to understand how these complicated molecules function in our bodies, in bacteria, and viruses. In many cases protein crystals are not of sufficient quality to enable determination of their structures using a technique called x-ray crystallography. Therefore, new applications that may improve the quality of protein crystals will be of significant benefit as scientists continue to try to understand the function of these complicated molecules.
Task Progress:	In the past year we finalized all the chemical conditions for the three proteins selected for our initial space station experiments. Crystals were grown in these capillaries using three different size proteins: 1) hen egg white lysozyme, 2) bovine serum albumin and 3) glutathione-s-transferase. Each protein solution contains single molecules of the specific protein (without any fluorescent label attached to the protein) mixed with a specific amount of the same protein existing as a fused aggregate. The fused aggregate was labeled with a molecule that fluoresces when viewed under a fluorescent microscope. By quantitative the intensity of fluorescence we can determine the percentage of aggregate that is incorporated into growing crystals. This enables us to determine how buoyancy-induced convection (caused by gravity forcing lighter molecules to move upward while heavier molecules flow downward) influences the percentage incorporation of the larger (and heavier) aggregated protein into the growing crystals. Three cassettes (each cassette containing eight crystallization capillaries) were filled with the protein and crystallization solutions followed by immediate cryopreservation at -80°C. The frozen cassettes were maintained at -80°C throughout the SPX 10 launch and transfer to the International Space Station (ISS). The first cassette was activated by removing it from the -80 freezer and placing it in the Light Microscopy Module (LMM) maintained at 22°C. The LMM fluorescent microscope was used to periodically observe the crystals throughout the growth phase of this experiment. The periodic observation allows the investigator team to measure crystal growth rates and microgravity as compared to those obtained in the ground control experiments. In addition, the fluorescence intensity provides a semi quantitative analysis of the amount of aggregated protein incorporated into each growing crystal. The first cassette was returned to Earth approximately one month later so that the crystals could be further analyzed using our laboratory fluorescent microscope and X-ray diffraction equipment. The goal of this experiment is to compare the percentage incorporation of aggregated protein in microgravity versus that due to gravity performed in our ground-based studies. Our initial analysis of the protein crystals return from the ISS suggest that there is a higher percentage of aggregate incorporated in the Earth control crystals compared to the microgravity-grown crystals. In addition, the initial growth rate analysis suggests that crystals grow more slowly in a microgravity environment compared to the Earth control crystals. Finally, although very preliminary, we also have observed improved microgravity crystal quality via x-ray diffraction analysis. NOTE: PI retired in 2017 from the University of Alabama, Birmingham, and is now Principal Scientist at The Aerospace Corporation, El Segundo, CA; project continues with same title as grant 80NSSC18K0013. See that project for subsequent reporting (Ed., 5/22/2018)
Bibliography Type:	Description: (Last Updated: 06/29/2023)